Example Reports in Public Health 2024; 19(3): 211–229

Exceedance probabilities as a basis for decisions in disease control

First Author[1], Second Author[2] and Third Author[1]

[1]Department of Example Studies, Example University [2]Institute of Placeholder Research

Abstract. Maps of predicted prevalence are often read as if they were certain. We argue that decisions about where to intervene should instead rest on the probability that prevalence exceeds a policy threshold, and we show how to compute and communicate this probability from a fitted geostatistical model. Workshops with programme staff suggested that three classes of exceedance probability were easier to act on than a continuous scale. We provide software and a template for reporting.

Keywords: decision making; exceedance probability; risk communication

1 Introduction

Disease control programmes plan their work district by district, but the data that inform those plans come from a limited number of survey locations. Between the locations we must predict, and every prediction carries uncertainty that should travel with it into the decision.

Geostatistical models treat the unobserved prevalence surface as a realisation of a stochastic process. The correlation between two locations decays with the distance between them at a rate set by the scale parameter of the correlation function. Locations close to a survey are predicted with more confidence than locations far from any survey, which is the behaviour a programme needs to see.

In this paper we describe the model, explain how it is fitted and checked, and show how its output can be summarised for decision makers. Section 2 introduces the data and Section 3 the model. Section 4 reports a simulation study, and Section 5 discusses limitations and extensions.

2 Data

The surveys recorded the number of people tested and the number who tested positive in each community, together with the geographical coordinates of the community and a set of environmental covariates derived from satellite images.

A mock first page made for the GPH slide template. The journal, authors and results are invented.